

Coarse-Grained ENM

- C_α virtual bonds & angles
- Elastic network ($R_{\text{cut}} = 8.5 \text{ \AA}$)
- Symplectic BAOAB ($\Delta t = 4.0 \text{ fs}$)
- Global domain vibrations

All-Atom Heavy Force Field

- Heavy atoms + AMBER ff14SB
- HCT Generalized Born + SASA
- Analytical Blondel-Karplus torques
- Metal coordination matrices

Funnel Metadynamics

- Radial CV $r = \|\mathbf{R}_L - \mathbf{R}_P\|$
- Flat-bottom bound restraint
- Well-tempered adaptive hills
- 3D metric Jacobian correction

Markovian Chemical Network & Transition Path Theory (TPT)

S_0 (Bulk) $\rightleftharpoons S_1$ (Encounter) $\rightleftharpoons S_2$ (Vestibule) $\rightleftharpoons S_3$ (Native Bound)

- Detailed balance: $\pi_i K_{ij} = \pi_j K_{ji}$ with chemical potential $\mu_{\text{bulk}} = G_0^\circ + k_B T \ln([L]/1 \text{ M})$
- Committed Dirichlet problem: $\sum_j K_{ij} q_j^+ = 0$ with boundary values $q_0^+ = 0$, $q_3^+ = 1$
- Chemical Master Equation $\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{K}$ & Recursive 1D birth-death MFPT ($\tau_{\text{MFPT}} = 317 \text{ } \mu\text{s}$)